

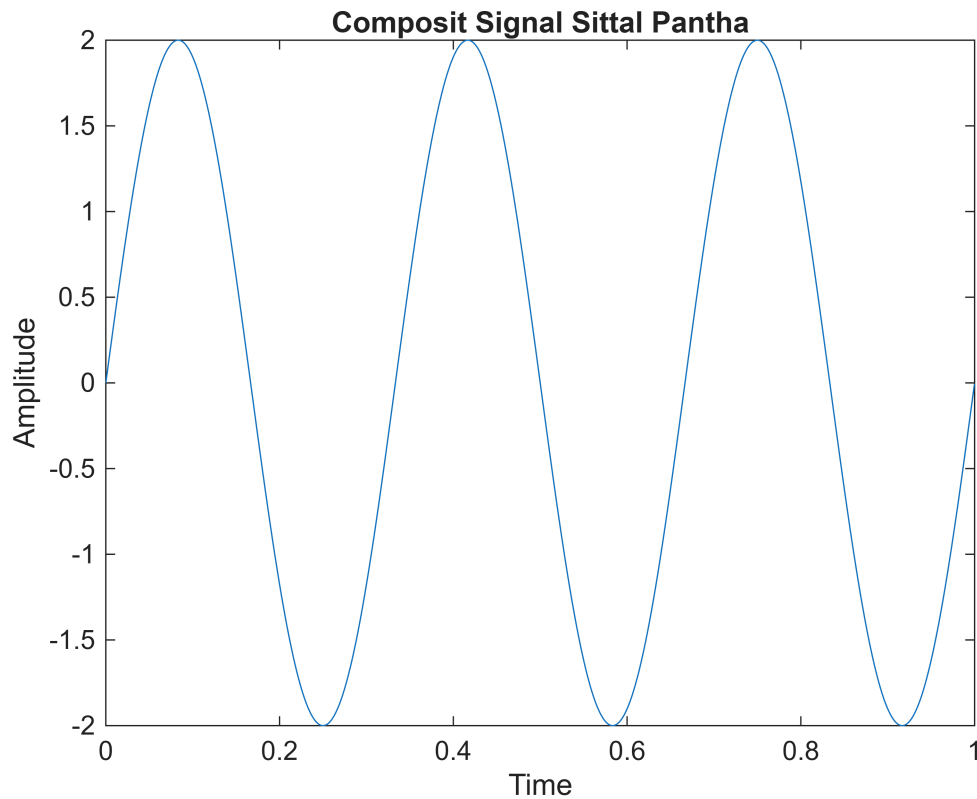
% Shannon Capacity Calculation for Noise Channel

```
clear all;
clc;

fs = 8000; % Sampling frequency
t = 0:1/fs : 1-1/fs; % Time duration
%cx = 1.1*sin(2*pi*100*t) + 1.3*cos(2*pi*300*t) + 1.5*sin(2*pi*2000*t);
cx = 2*sin(2*pi*3*t);

plot(t,cx);
xlabel('Time')
ylabel('Amplitude')

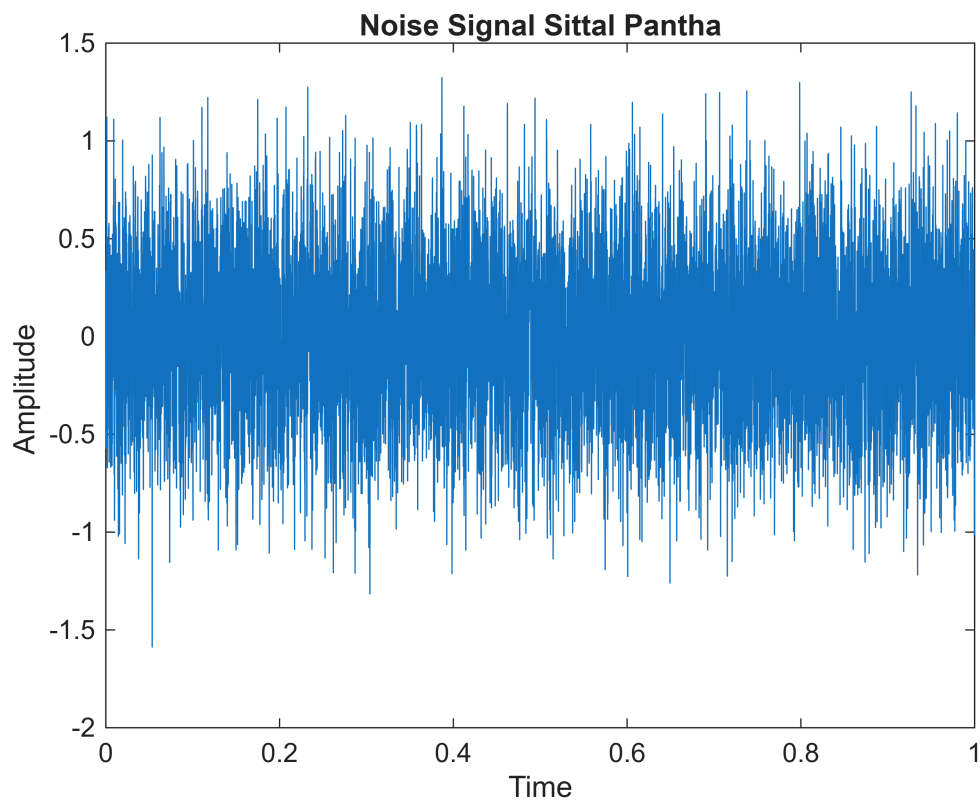
title('Composit Signal Sittal Pantha');
```



```
ns = 0.4 *randn(size(cx));

plot(t,ns);
xlabel('Time')
ylabel('Amplitude ')

title('Noise Signal Sittal Pantha');
```

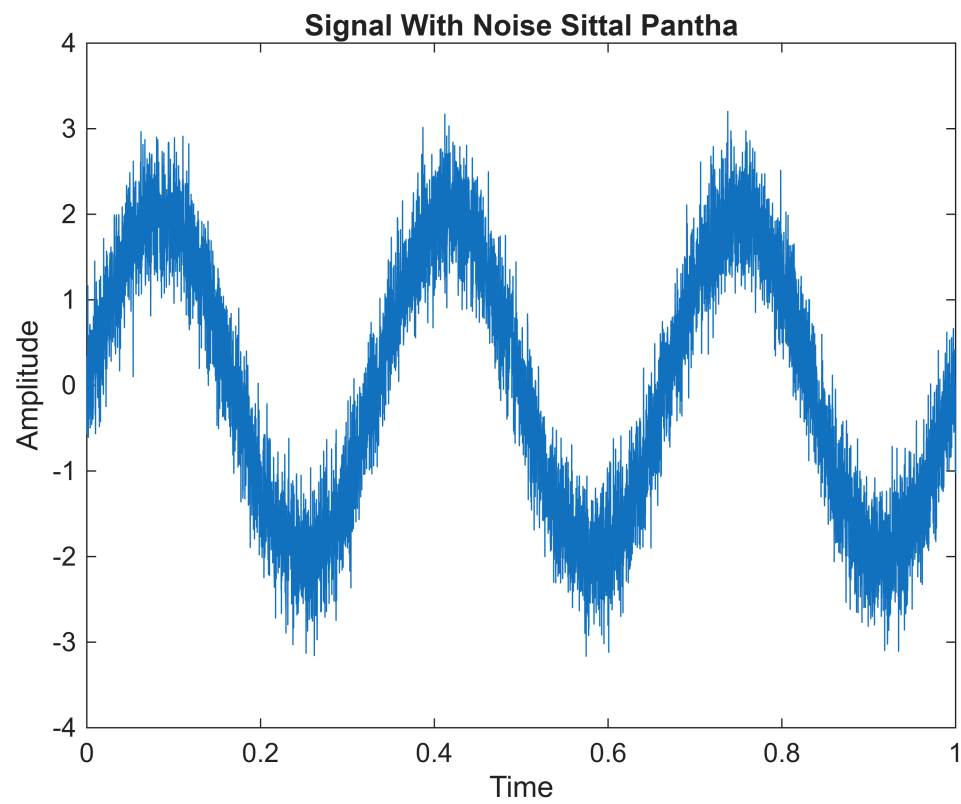


```
% signal with noise

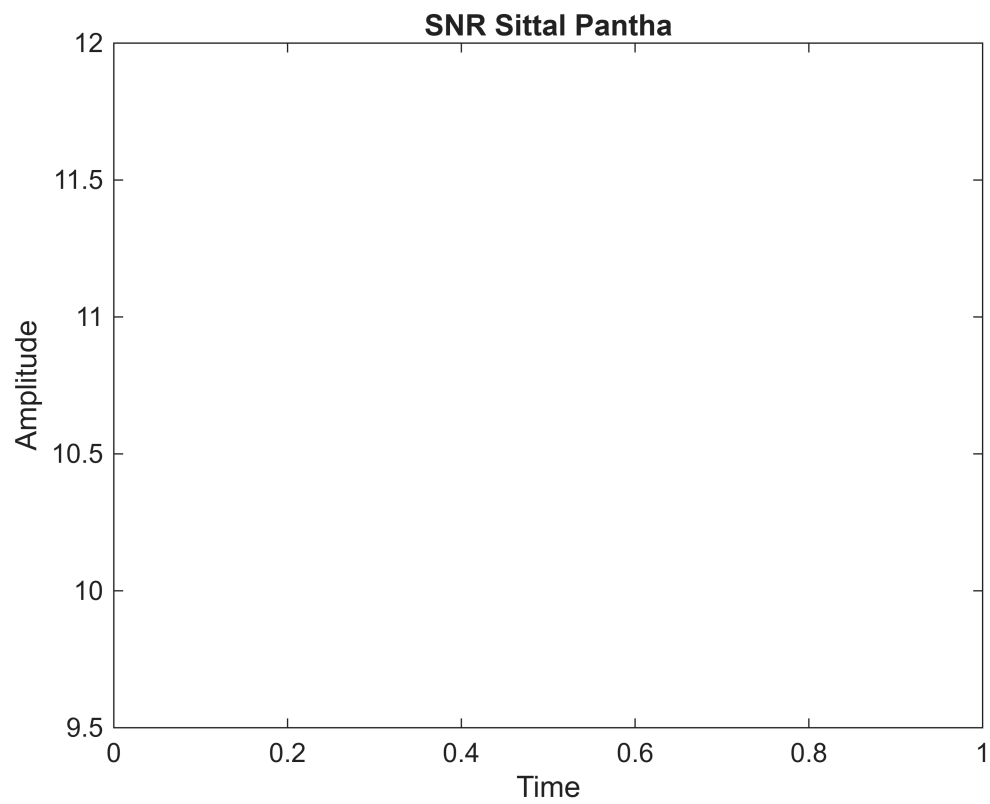
S_W_N = cx + ns;

plot(t,S_W_N);
xlabel('Time')
ylabel('Amplitude')

title('Signal With Noise Sittal Pantha');
```



```
S_N_R = snr(S_W_N,fs);  
  
plot(t,S_N_R);  
xlabel('Time')  
ylabel('Amplitude')  
  
title('SNR Sittal Pantha');
```



```
bandwidth = obw(cx,fs); % Bandwidth of the signal  
L=2; % Level of the signal  
C = bandwidth*log2(1+S_N_R)
```

```
C =  
3.5457
```